

SLEEP METHOD

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- The sleep() method in Java **pauses** a thread for a specific time. After that, the thread **resumes execution**.

We have 2 sleep method :

1. Public static native void sleep(long milliseconds) -> remember that native methods are those method who implementation is in different language.
 - It will throw compile time exception that is -> InterruptedException.
2. Public static void sleep(long milliseconds , int nano)
 - It will throw compile time exception that is -> InterruptedException.

Basics of sleep()

1. **Pauses a thread for a given time** (milliseconds).
2. **Does not release the lock** (if holding one).

Syntax:

Thread.sleep(milliseconds);

- **Real-Life Example: Alarm Clock** 🕒

Real-World Scenario Java sleep()

You set an alarm for 5 minutes Thread.sleep(300000); // 5 minutes

Alarm pauses but stays active **Thread pauses but doesn't stop**

Alarm rings after 5 minutes **Thread resumes after sleep time**

Just like an **alarm waits for the set time** before ringing, sleep() **makes the thread wait** before continuing execution.

CODE:

```
package multithreading;
class MyThread2 extends Thread {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            try {
                Thread.sleep(1000); // Pause for 1 second
                System.out.println(Thread.currentThread().getName() + " - Count: " + i);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}

public class SleepExample {
    public static void main(String[] args) {
        MyThread2 t1 = new MyThread2();
        MyThread2 t2 = new MyThread2();

        t1.setName("Thread 1");
        t2.setName("Thread 2");

        t1.start();
        t2.start();
    }
}
```

- Build a timer watch using this :

```
package multithreading;
class TimerWatch extends Thread {
    public void run() {
        int minutes = 0;
        int seconds = 0;

        while (true) {
            System.out.printf("%02d:%02d\n", minutes, seconds); // Display time in MM:SS format
            try {
                Thread.sleep(1000); // Pause for 1 second
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}
```

```

        // Increment time
        seconds++;
        if (seconds == 60) { // If seconds reaches 60, reset and increase minutes
            seconds = 0;
            minutes++;
        }
    }
}

public class TimerWatchExample {
    public static void main(String[] args) {
        TimerWatch timer = new TimerWatch();
        timer.start(); // Start the timer thread
    }
}

```

sleep() method important points

1. If the value of milliseconds is negative then "IllegalArgumentException" is thrown.
2. If the value of nanoseconds is not in the range 0-999999 then "IllegalArgumentException" is thrown.
3. Whenever we want to use the sleep() method we also need to handle the "InterruptedException".
If we will not handle it, the JVM will show a compilation error.
4. When any thread is sleeping and if any other thread interrupts it, then it throws "InterruptedException".
5. The sleep() method always pauses the current thread execution. When the JVM finds the sleep() method in code, it checks that which thread is running and pause the execution of thread.
6. When we use sleep() method to pause the execution of thread. the thread scheduler assigns the CPU to another thread if any thread exists. So, there is no guarantee that the thread wakes up exactly after the time specified in sleep() method. It totally depends on the thread scheduler.
7. While the thread is sleeping, it doesn't lose any locks or monitors that it had acquired before sleeping.